FCG-DETR: Foreground Confidence-Guided Detection Transformer for Remote Sensing Oriented Object Detection

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***Abstract*—Accurate detection of oriented objects in remote sensing imagery is often hindered by various aspect ratios and complex background interference. While Transformer-based detectors establish an end-to-end pipeline by eliminating nonmaximum suppression (NMS), they incur substantial computational overhead due to indiscriminate processing of all tokens. This limitation not only leads to unnecessary resource consumption but also compromises detection performance. To address this issue, we propose FCG-DETR, an efficient-oriented object detection framework that leverages hierarchical foreground guidance, dynamic token selection, and confidence-gated decoding to reduce computational overhead while improving detection accuracy. Specifically, the hierarchical foreground generation (HFG) module leverages anisotropic Gaussian modeling to generate category-agnostic spatial priors, which serve as soft-supervision signals to guide the encoder toward high potential regions across varying scales and aspect ratios. To further optimize efficiency, we design a dynamic foreground token selection (DFS) strategy. Instead of uniform sampling, DFS utilizes normalized cross-scale metrics to adaptively align token distribution with the object density of each feature level. Finally, a foreground confidence-gated attention (FCGA) mechanism is proposed to resolve the optimization conflict in the decoder. By dynamically modulating the reliance on spatial priors based on query confidence, FCGA enables precise refinement for converged queries while ensuring robust exploration for uncertain ones. Extensive experiments on DOTA and DIOR-R benchmarks demonstrate that FCG-DETR achieves state-of-the-art performance with faster convergence and significantly reduced computational overhead. The code is available at https://github.com/gyy1502/FCG-DETR**

***Index Terms*—Confidence-gated attention, DETR, Gaussian modeling, oriented object detection, token selection.**

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I. Introduction

ARBITRARY-ORIENTED object detection (AOOD) has been widely applied in tasks such as scene text detection and remote sensing object detection. Unlike natural images, remote sensing imagery presents unique challenges: objects exhibit significant variations in aspect ratios and scales, appear in arbitrary orientations, and are often embedded in complex, large-area background scenarios [1]. Traditional horizontal object detection methods, which rely on horizontal bounding boxes, often introduce excessive background noise and struggle to separate densely packed targets effectively. In contrast, by employing oriented bounding boxes (OBBs) [2], AOOD can more accurately capture the spatial extent of targets, providing a robust solution for aerial interpretation.

Benefiting from deep learning, CNN-based detectors have achieved remarkable progress. Among these methods, some first generate high-quality candidate regions, and then perform further classification and regression on these regions [3], [4], [5], [6], while others directly predict the bounding box and orientation in one step [7], [8], [9]. They usually rely on nonmaximum suppression (NMS) to eliminate redundant detection boxes. However, this operation introduces substantial computational overhead in dense scenarios, thereby compromising inference efficiency.

DETR addressed this limitation by introducing an NMS-free end-to-end pipeline via bipartite matching [10]. Following the success of numerous optimized variants [11], [12], [13], [14], this paradigm has been successfully adapted for oriented object detection [15], [16], [17], [18]. However, standard global attention scales quadratically with image size, making it computationally costly and highly inefficient for high-resolution aerial imagery. Consequently, researchers have turned to sparse attention mechanisms. Initial efforts, such as Deformable DETR [13], introduced sparse sampling strategies that restrict attention to a small set of learned key points, thereby avoiding full-map scanning. Building on this, subsequent works like Sparse DETR [19], [20], [21] have further mitigated computational overhead by employing scoring mechanisms for token pruning. Similarly, in the remote sensing domain, methods like SANet [22] explore patch-level filtering to exclude background redundancy. However, their strategies predominantly rely on axis-aligned priors designed for horizontal objects. Such

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modeling approaches present inherent limitations when applied to remote sensing targets, which exhibit arbitrary orientations and various aspect ratios (e.g., slender bridges or diagonal ships). Consequently, they often struggle to fully align with the anisotropic geometric properties (i.e., objects with directional dependencies and nonuniform spatial characteristics) of aerial targets, leading to imprecise token selection that either misses critical object details or retains excessive background redundancy.

Furthermore, how to effectively select these tokens across multiscale features constitutes another pivotal factor influencing detection accuracy. The majority of existing sparse methods adopt a static select strategy, which enforces a fixed pruning ratio or token count across all feature levels, regardless of scene content. Although dynamic methods like ATS-DETR [23], Adaptive DETR [24] and the recent HyDeTr [25] attempt to adjust token allocation based on image complexity or density, they fail to achieve an optimal cross-scale balance. ATS-DETR performs token selection on the concatenated multiscale sequence, where high saliency scores tend to overshadow the lower scores of small objects. Adaptive DETR computes levelwise weights independently. Without interlevel competition, this leads to imbalances that overload shallow layers with redundancy while leaving high-level semantic information insufficient. Similarly, although HyDeTr [25] incorporates density guidance, it primarily targets query initialization for the decoder, failing to address the substantial computational redundancy inherent in processing dense feature maps within the encoder. Remote sensing scenes typically feature both sparse large objects and dense small clusters, presenting a dual challenge. For small targets, detection is hindered by limited spatial extents and weak feature responses [26], [27]. Conversely, for large targets, expansive receptive fields in deep layers often waste computational resources on redundant backgrounds. Current methods overlook this heterogeneity, leading to imbalanced token allocation. To address this, a density-aware mechanism is required to dynamically shift computational resources to regions with high target density.

To accelerate convergence, DETR architectures have transitioned from random query initialization to leveraging explicit spatial priors, such as reference points or anchor boxes. SMCA [28] leverages a Gaussian-like weight mechanism to model and compute spatial attention; in contrast, DAB-DETR [29] and Conditional DETR [30] adopt bounding boxes as carriers to explicitly encode the positional attributes of object queries. DN-DETR [31] and DINO [12] have reconstructed ground truth from noisy boxes, which effectively stabilizes bipartite matching and significantly boosts convergence efficiency. However, the decoder employs a uniform processing strategy across all object queries, overlooking differences in their semantic reliability and convergence states. This lack of differentiation limits the model's ability to learn optimal parameters.

To address these challenges, we propose FCG-DETR, a novel foreground-aware framework tailored for background-dominated remote sensing imagery. As visualized in Fig. 1, by integrating geometry-aware foreground modeling, adaptive token selection, and confidence-gated decoding, our method strictly filters redundant background information while retaining salient target features. Consequently, it achieves a superior trade-off between computational efficiency and detection accuracy in complex high-resolution scenarios. The main contributions of this article are summarized as follows.

1) We design a hierarchical foreground generation (HFG) module to construct geometry-aware spatial priors. Leveraging anisotropic Gaussian modeling and hierarchical propagation, HFG generates adaptive, class-agnostic confidence maps that effectively guide the encoder to suppress background noise.
2) We develop a dynamic foreground token selection (DFS) strategy to optimize cross-scale feature sampling. Instead of uniform allocation, DFS utilizes normalized cross-scale metrics to adaptively distribute tokens according to target density, ensuring efficient information preservation across varying scales.
3) We introduce a foreground confidence-gated attention (FCGA) mechanism to resolve the optimization conflict between converged and unconverged queries. By dynamically modulating the reliance on spatial priors based on query confidence, FCGA allows high confidence queries to leverage spatial priors for precise localization while ensuring that uncertain queries maintain flexibility for robust updating.
4) Extensive experiments on the DOTA and DIOR-R datasets validate the superiority of our approach. The results demonstrate that FCG-DETR achieves state-of-the-art performance, significantly outperforming existing methods in terms of both detection accuracy and computational efficiency.

II. Related Work

A. Arbitrary-Oriented Object Detection

Benefiting from the emergence of large-scale open-source datasets, the task of AOOD has witnessed rapid development. Currently, mainstream detectors predominantly adopt convolutional neural network (CNN) frameworks, which are generally derived from classical horizontal detectors by introducing OBB regression. Representative methods, such as Rotated Faster R-CNN [4], FCOSR [3], SCRDet++ [5], R3Det [8], and S2A-Net [32], incorporate an angle parameter θ into the prediction targets, achieving impressive performance through refined feature alignment. To mitigate the discontinuity and ambiguity issues inherent in angle regression, advanced encoding schemes like circular smooth label (CSL) [33], KLD [34], and PSC [35] have been proposed to enable continuous and boundary-aware angle prediction. Meanwhile, from the perspective of inference efficiency, one-stage detectors such as the YOLO series (e.g., YOLOv5 and YOLOv8), RTMDet [9], and BiFA-YOLO [7] have become practical choices for real-time applications due to their hardware-friendly architectures. However, CNN-based methods typically rely on NMS for postprocessing. While NMS removes redundant boxes, it slows inference, relies on sensitive hyperparameters, and often causes missed detections in complex remote sensing scenes.

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Additionally, limited by local receptive fields, conventional CNNs fail to capture global dependencies, struggling to represent targets with varied scales, arbitrary orientations, and dense distributions.

B. Transformer-Based Object Detection

To address these limitations, Transformer-based detection methods have emerged. DETR established a new paradigm by framing object detection as a direct set prediction task. It utilizes a Transformer encoder–decoder architecture and bipartite matching to achieve one-to-one label assignment, effectively eliminating the need for NMS. However, the original DETR suffers from slow convergence and high computational complexity. To address these limitations, Deformable DETR [13] introduced multiscale deformable attention (MSDA) to restrict computation to sparse sampling points. Subsequent works like Condition-DETR [14] and DAB-DETR [29] further accelerated convergence by introducing conditional spatial queries and explicit 4-D box coordinates as query anchors.

Leveraging these developments, numerous studies have adapted Transformers for oriented object detection in remote sensing. O2DETR [36] marked the first attempt to design a Transformer-based rotation detector. To enhance orientation perception, AO2-DETR [15] introduced an oriented proposal generation mechanism based on Deformable DETR, while OrientFormer [37] incorporated Gaussian encoding to capture directional information. RotaTR [38] designed a rotation-sensitive attention module to extract rotation-equivariant features. Similarly, FPNTransformer [39] devised top-down and bottom-up cross-scale transformer decoders to achieve both rotation-invariant and rotation-equivariant modeling, effectively handling objects with drastic orientation and scale variations. More recently, ARS-DETR [17] proposed an aspect ratio-aware circle smooth label to handle the boundary ambiguity of angles. OrientedDINO [18] further pushed the state-of-the-art by decoupling angle prediction from box refinement, significantly improving detection accuracy.

Among the aforementioned methods, the computational complexity of the attention mechanism increases quadratically with the number of tokens. Even with deformable attention, the number of tokens or sampling points remains substantial. Crucially, objects in aerial images typically occupy only a small fraction of the spatial domain. This observation suggests that eliminating background tokens from multiscale feature maps could significantly reduce overhead without compromising performance, motivating the sparse strategy proposed in this work.

C. Sparsity and Convergence in Detection Transformer

To alleviate the issue of high computational complexity, early research primarily focused on two directions: restricting the attention scope and optimizing the model architecture. Among them, building upon Deformable DETR, Efficient-DETR [40], and Lite-DETR [41] further optimized the encoder–decoder structure. By prioritizing the update of high-level features, they minimized redundant computations to the greatest extent.

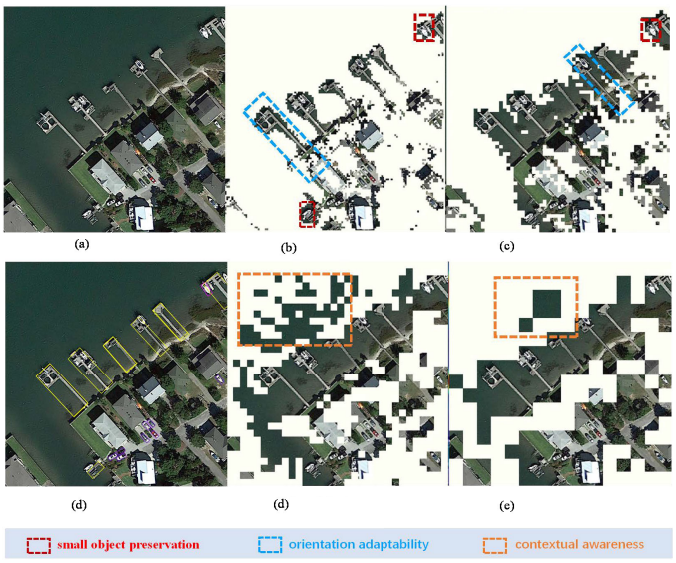


Fig. 1. Visualization of the selected tokens in our model. At lower levels (b) and (c), it accurately preserves small objects (red boxes) and adapts to the arbitrary orientations of elongated structures like the harbor (blue box), effectively filtering out background noise. At higher levels (d) and (e), it shifts to semantic regions that embrace broader context (orange box), such as the surrounding water, furnishing the detector with rich features. (a) Image. (b) Level 1 tokens. (c) Level 2 tokens. (d) Output. (d) Level 3 tokens. (e) Level 4 tokens.

Beyond structural optimization, token pruning strategies have been explored to actively discard uninformative features. Sparse DETR [19] and Focus-DETR [20] introduce a scoring network to select only the top-k% tokens with the highest saliency for the encoder, significantly reducing the token numbers. Similarly, PnP-DETR [42] abstracts the background into coarse-grained vectors to reduce complexity. However, a critical limitation of these methods is their reliance on static selection ratios. They fail to adapt to the varying foreground sparsity across different images, discarding valid targets in dense scenes while retaining redundancy in sparse ones.

In parallel, optimizing the decoder to expedite convergence has attracted significant attention. Early works like SMCA and DN-DETR focus on enhancing spatial priors or stabilizing bipartite matching via denoising tasks. Beyond query formulation, optimizing the supervision strategy has proven effective. Methods such as Group-DETR [11] and Hybrid-DETR [43] introduce auxiliary branches with diversified positive samples to boost feature learning, while MS-DETR [44] directly imposes mixed supervision on the primary decoder. However, whether refining priors or enriching supervision, these methods fundamentally overlook the discrepancies in query states. This uniform processing mechanism fails to reconcile the divergent optimization requirements of different queries, thereby becoming a bottleneck that restricts the improvement of model performance.

III. Proposed Method

The overall architecture of FCG-DETR is illustrated in Fig. 2. Our framework is fundamentally designed around the explicit utilization of foreground priors to optimize DETR-based detection in remote sensing imagery. Specifically, we

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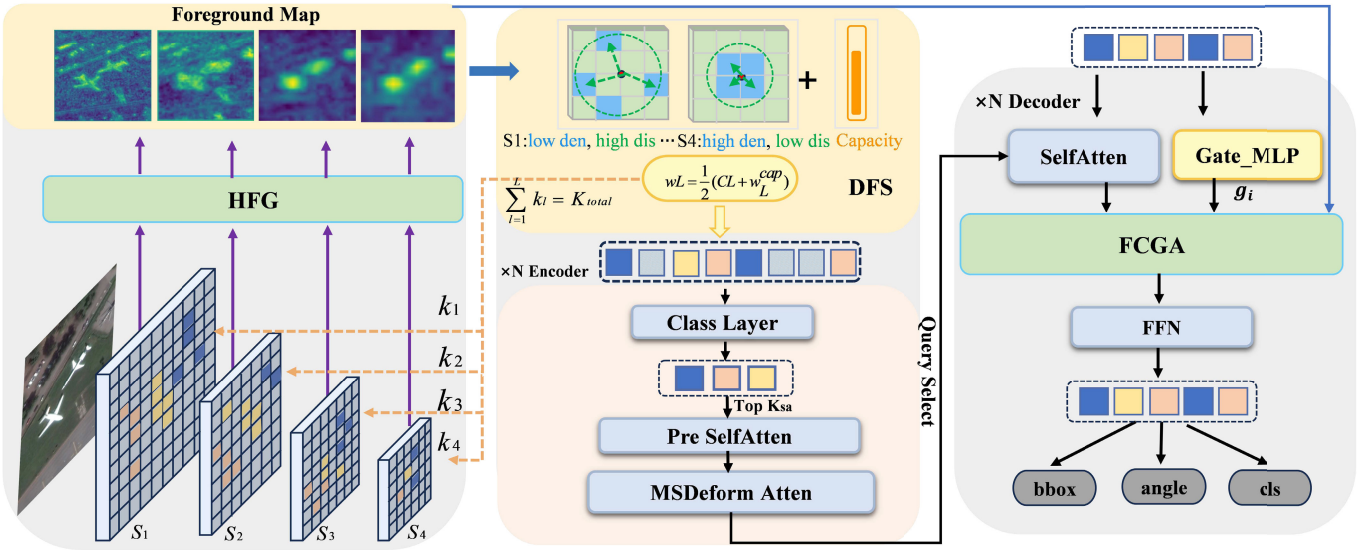


Fig. 2. Overall architecture of the FCG-DETR. The framework features a tightly coupled pipeline centered on foreground-guided feature refinement. First, the HFG module extracts category-agnostic score maps from multilevel features as unified spatial priors. Guided by these maps, the encoder uses DFS to discard background redundancy, retaining only informative foreground tokens. Finally, the decoder processes the selected tokens and HFG score maps via FCGA, which leverages the spatial priors to adaptively modulate query attention based on semantic certainty, accelerating convergence.

construct an HFG module, as detailed in Section III-A. Guided by these priors, the input tokens for the encoder are adaptively selected to reduce redundancy, with the specific mechanism described in Section III-B. Subsequently, Section III-C presents a confidence-gated mechanism to resolve the optimization conflict in the decoder. By dynamically modulating the spatial constraints in cross-attention, this module ensures that high-confidence queries focus on fine-grained feature extraction around their reference points, while low-confidence queries retain broader receptive fields to correct potential localization errors, thereby significantly improving convergence efficiency.

A. Hierarchical Foreground Generation

In the processing of high-resolution remote sensing images, directly feeding the multiscale feature maps $\{f_i\}_{i=1}^{L}$ (where $L = 4$ and $f_l \in \mathbb{R}^{C\times H_l\times W_l}$) from the backbone network into the transformer encoder poses two major challenges. Primarily, the large feature dimensions lead to prohibitive computational complexity. Concurrently, the abundance of redundant background features weakens the attention mechanism's capacity to discriminate salient targets from irrelevant information. To address these issues, it is crucial to furnish the encoder with a robust, category-agnostic spatial prior, which guides the allocation of computational resources toward regions with high target potential.

The HFG module constructs an adaptive, class-agnostic foreground confidence map to accommodate varying target scales, orientations, and aspect ratios. Acting as a differentiable soft-supervision signal, this map provides geometry-aware spatial priors that strictly guide the Transformer encoder to prioritize high-potential regions. This is achieved in two stages: first, anisotropic Gaussian models generate continuous pseudo-labels; subsequently, a multiscale foreground extraction head hierarchically predicts the confidence maps supervised by these labels. Fundamentally distinct from conventional methods that utilize Gaussians merely as regression metrics (e.g., KLD [34]) or rely on rigid binary assignments, our continuous formulation explicitly prevents tiny instances from being submerged by boundary noise.

1) Foreground Prior Generation: To accurately model remote sensing targets with arbitrary orientations and scales, the foreground probability of any given image coordinate $\mathbf{c} = (x, y)$ relative to ground-truth rotated bounding box $\mathcal{B}_{\text{rot}}$ is formulated using a 2-D anisotropic Gaussian distribution.

First, to eliminate the impact of rotation and scale variations, the global coordinate $\mathbf{c}$ is transformed into a local coordinate system $\mathbf{c}' = (\mathbf{c}'_x, \mathbf{c}'_y)$. This system is centered at the midpoint μ of $\mathcal{B}_{\text{rot}}$ and aligned with the principal axes of $\mathcal{B}_{\text{rot}}$, as shown in

$$\mathbf{c}' = \left(\mathbf{c}'_x, \mathbf{c}'_y\right) = \mathbf{R}_\theta^\top (\mathbf{c} - \boldsymbol{\mu}) \tag{1}$$

where μ and θ denote the center and orientation angle of $\mathcal{B}_{\text{rot}}$, and $\mathbf{R}_\theta$ represents the standard 2-D rotation matrix

$$\mathbf{R}_\theta = \begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix}. \tag{2}$$

A normalized elliptical distance $d_{\text{norm}}(\mathbf{c})$ is then defined to measure the relative distance from $\mathbf{c}$ to the center of $\mathcal{B}_{\text{rot}}$. Normalization is implemented via the half-width σ_w and half-height σ_h of $\mathcal{B}_{\text{rot}}$. Its computational formula is given by

$$d_{\text{norm}}(\mathbf{c}) = \sqrt{\frac{\left(c'_x\right)^2}{\sigma_w^2} + \frac{\left(c'_y\right)^2}{\sigma_h^2}}. \tag{3}$$

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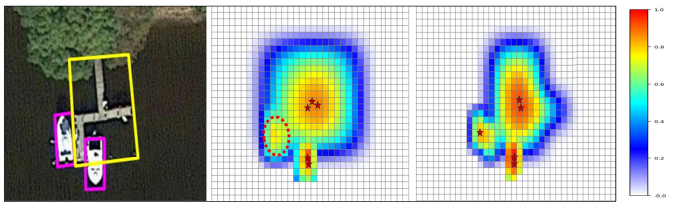


Fig. 3. Visualization of foreground pseudo-label generation for (a) original image with ground-truth bounding boxes of small vehicles and harbors, using (b) horizontal and (c) rotated anisotropic Gaussian modeling. The heatmap represents normalized weight scores ranging from 0 to 1, where blue indicates lower values and red indicates higher values. Asterisk markers indicate the top-5 candidate queries selected based on foreground confidence.

Subsequently, $d_{\text{norm}}(\mathbf{c})$ is mapped to a foreground confidence score $S(\mathbf{c}, \mathcal{B}_{\text{rot}})$ using an unnormalized Gaussian function

$$S(\mathbf{c}, \mathcal{B}_{\text{rot}}) = \begin{cases} \exp\left(-\frac{1}{2}(g \cdot d_{\text{norm}}(\mathbf{c}))^2\right), & \text{if } d_{\text{norm}}(\mathbf{c}) \le 1 \\ 0, & \text{if } d_{\text{norm}}(\mathbf{c}) > 1. \end{cases} \tag{4}$$

The confidence score map composed of $S(\mathbf{c}, \mathcal{B}_{\text{rot}})$ serves as the ground-truth supervision for our foreground prediction head. The hyperparameter g controls the decay rate of the Gaussian function, typically set to 2 in our experiments; it ensures that high confidence scores are strictly confined to the most discriminative central regions. This formulation produces a score that achieves its maximum value of 1 at the target center ($d_{\text{norm}} = 0$) and decays smoothly to its boundary ($d_{\text{norm}} = 1$), while remaining zero outside. To handle multiple valid targets within the same feature level, each grid point adopts the maximum confidence score across all overlapping targets as its final foreground confidence. This max-pooling strategy effectively resolves spatial ambiguities while preserving the most salient foreground characteristics, thereby generating a smooth foreground confidence map for the current feature level.

2) Hierarchical Foreground Learning: Inspired by Focus-DETR [20], we utilize a hierarchical propagation mechanism operating top-down across multiple feature pyramid levels to propagate foreground confidence maps. Unlike FPNs that propagate abstract semantic vectors, and Focus-DETR that relies on rigid binary (0/1) assignments, our approach explicitly propagates continuous 2-D geometric probabilities. Governed by anisotropic Gaussian modeling, the propagated map S_l inherently encodes shape, scale, and orientation priors. Consequently, moving beyond simple semantic fusion or mere feature enhancement, this mechanism acts as a strict progressive spatial filter. It utilizes high-confidence, structurally constrained probabilities from deeper layers to systematically prune false positives and boundary noise in shallower layers. Specifically, for each pyramid level l, the foreground confidence map S_l from the higher layer is upsampled and used to modulate the current feature map f_{l-1}. This cross-level modulation enhances foreground-relevant features while suppressing background noise, formulated as follows:

$$S_{l-1} = \mathbf{MLP_F}\left(f_{l-1} \odot (1 + \mathbf{UP}(\alpha_l \cdot S_l))\right). \tag{5}$$

Here, S_l denotes the foreground confidence map of the lth layer, **UP** represents bilinear upsampling, and $\mathbf{MLP_F}$ is a lightweight foreground predictor. Specifically, MLP_F shares weights across layers. It first projects the input to a hidden dimension H (typically 256) via LayerNorm, a Linear layer, and a GELU activation. For local–global fusion, local token features are concatenated with their global context captured via global average pooling (GAP). Finally, three successive Linear-GELU blocks progressively compress the channel dimension ($H \to H/2 \to H/4 \to 1$). Additionally, the learnable modulation coefficients $\{\alpha_l\}_{l=2}^{4}$ are initialized from a uniform distribution $\mathcal{U}[-0.1, 0.1]$. The training of $\mathbf{MLP_F}$ is supervised by the pseudo-labels generated in the first stage through focal loss, which effectively handles foreground–background class imbalance. As illustrated in Fig. 3, rotated modeling adapts to target orientation and geometry, producing tighter, more accurate foreground regions compared to axis-aligned horizontal modeling, which helps preserve small target discrimination during subsequent top-k sampling.

B. Dynamic Foreground Token Selection

While DETR-based encoders require a fixed number of input tokens for computational efficiency, existing methods that uniformly allocate tokens across feature levels fail to adapt to varying target density distributions at different scales. We observe that the foreground confidence map implicitly encodes density information alongside positional priors and confidence estimates. This naturally motivates us to propose the DFS module, which adaptively distributes the fixed token numbers based on density analysis. Specifically, DFS first computes normalized cross-scale metrics to evaluate each feature level's information content, then adaptively allocates tokens across levels based on these metrics, and finally performs top-k selection within each level.

1) Normalized Cross-Scale Metrics: Unlike prior methods that rely on absolute variance statistics, which inherently favor large-scale feature maps due to their greater number of pixels, we introduce two normalized metrics to enable fair cross-scale comparison. For each feature level L with foreground probability map $p_L \in \mathbb{R}^{H_L \times W_L}$, we compute the following.

Information Density: We define the information density as the confidence-weighted average information per valid pixel

$$\mathcal{I}_L = \frac{1}{N_L^{\text{valid}}} \sum_{i=1}^{N_L} \left[p_L(i) \cdot c_L(i) \cdot \mathbb{I}_{\text{valid}}(i)\right] \tag{6}$$

where $c_L(i) = 2|p_L(i) - 0.5|$ is the confidence weight that assigns higher importance to predictions closer to 0 or 1, N_L^{valid} is the number of valid (nonmasked) pixels, and $\mathbb{I}_{\text{valid}}(i)$ is the validity indicator. Normalizing by pixel count ensures that layers with different resolutions are comparable, a 128×128 map is not automatically favored over a 16×16 map simply due to having more pixels.

Relative Dispersion: To quantify the diversity of information distribution within each layer, we compute the relative dispersion, defined as the ratio of standard deviation to mean

$$\mathcal{D}_L = \frac{\sigma_L}{\mu_L + \epsilon} \tag{7}$$

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where μ_L and σ_L are the mean and standard deviation of valid foreground probabilities in level L, and ϵ is a small constant for numerical stability. This scale-invariant metric captures whether a layer contains uniform responses (low $\mathcal{D}_L$) or heterogeneous patterns such as dense object clusters or sharp boundaries (high $\mathcal{D}_L$). Unlike kurtosis which primarily detects extreme outliers, relative dispersion provides a more robust measure of structural complexity across scales. As shown in Fig. 2, the shallow layer (S1) exhibits low density and high dispersion, indicating that it primarily captures object edges. Although less informative, these features are crucial for localization. In contrast, the deep layer (S4) presents high density and low dispersion, denoting that it captures highly informative, spatially concentrated semantic features.

2) Adaptive Token Allocation Strategy: To determine the optimal token numbers for each layer, we combine content-aware metrics with capacity constraints through a principled fusion strategy. We first normalize both information density and relative dispersion across layers to form probability distributions

$$\tilde{\mathcal{I}}_L = \frac{\mathcal{I}_L}{\sum_{l=1}^{L_{\text{total}}} \mathcal{I}_l}, \quad \tilde{\mathcal{D}}_L = \frac{\mathcal{D}_L}{\sum_{l=1}^{L_{\text{total}}} \mathcal{D}_l}. \tag{8}$$

These normalized metrics are then fused with manually tuned weight coefficients to form a content score

$$\mathcal{C}_L = w_{\mathcal{I}} \cdot \tilde{\mathcal{I}}_L + w_{\mathcal{D}} \cdot \tilde{\mathcal{D}}_L. \tag{9}$$

The higher weight on relative dispersion emphasizes layers with complex structure, while information density prevents neglecting high-quality features. However, content-aware metrics alone may overallocate tokens to high resolution layers. To address this, we introduce a logarithmic capacity weighting mechanism. Directly using linear capacity ratios $S_L / \sum S_l$ (where $S_L = H_L \times W_L$) would allocate 75% of tokens to the largest layer, undermining content adaptivity. Instead, we apply logarithmic scaling

$$w_L^{\text{cap}} = \frac{\log(S_L + 1)}{\sum_{l=1}^{L_{\text{total}}} \log(S_l + 1)}. \tag{10}$$

This transformation converts the heavily skewed distribution into a more balanced one, allowing smaller layers to compete fairly while still respecting their representational capacity. The final allocation weight combines content awareness and capacity constraints with equal importance

$$\mathcal{W}_L = \frac{1}{2}\left(\mathcal{C}_L + w_L^{\text{cap}}\right). \tag{11}$$

The initial token allocation for each layer is then computed as follows:

$$k_L^{\text{init}} = \text{round}\left(K_{\text{total}} \times \mathcal{W}_L\right) \tag{12}$$

where K_{total} is the total token number. To ensure $\sum_l k_l = K_{\text{total}}$ exactly, we apply a refinement procedure: if the sum falls short, we proportionally add tokens to layers with high composite content scores $\mathcal{C}_L$ and remaining capacity; if it exceeds the budget, we remove tokens from layers with low content scores

$$\mathcal{T}_L = \text{TopK}\left(s_L, k_L\right) \tag{13}$$

Algorithm 1 DFS for Encoder

Input: Multi-scale features $\{F_L\}$, Foreground scores $\{S_L\}$, Classifier W_{cls}, Total budget K_{total}, Number of pre-attention tokens K_{sa}, Layer-wise token retention N_{i+1}
Output: Final queries Q_{out}

Stage 1: Adaptive Token Allocation
for each level L **do**
  Compute $\mathcal{I}_L$, $\mathcal{D}_L$, allocation weight $\mathcal{W}_L$
  $k_L \leftarrow \text{round}(K_{\text{total}} \times \mathcal{W}_L)$
  $Q_L, P_L \leftarrow \text{TopK}(F_L, S_L, k_L)$
end for
$Q_0 \leftarrow \text{Concat}(\{Q_L\})$, $P_0 \leftarrow \text{Concat}(\{P_L\})$
Stage 2: Two-Stage Encoder Processing
for each encoder layer i **do**
Semantic score: $S_{sem} \leftarrow \max(W_{\text{cls}} \cdot Q_i)$
  Fuse with HFG map: $S^c \leftarrow S_{sem} \odot S_L$
  Select top-K_{sa}: idx $\leftarrow \text{TopK}(S, K_{sa})$
  $Q_i' \leftarrow \text{MHA}(Q_i[\text{idx}])$ and scatter back
  $Q_i'' \leftarrow \text{MSDeformAttn}(Q_i', \{F_L\})$
  $Q_i'' \leftarrow \text{FFN}(Q_i'')$
  if not final layer **then**
    Refine tokens: $Q_{i+1} \leftarrow \text{TopK}(Q_i'', N_{i+1})$ based on updated scores
  end if
end for
return Q_{out}

where s_L denotes the foreground probability at level L. This ensures that within each layer's allocated budget, only the most informative regions are attended to.

3) Integration With Encoder: Based on the token allocation from DFS, the encoder employs a two-stage attention structure for efficient processing. First, self-attention (MHA) performs dense relation modeling on the K_{sa} highest-scoring queries per level, enhancing features of high-confidence regions. Subsequently, cross-level feature interaction is implemented via MSDA, enabling all selected tokens to aggregate contextual information from multiple scales and thus accomplish global enhancement and enrichment of features. Compared to processing all feature map tokens (complexity $\mathcal{O}(N^2)$), this design reduces computational complexity to $\mathcal{O}(K_{\text{total}}^2)$, significantly improving small object detection accuracy while controlling computational cost. Algorithm 1 summarizes our complete pipeline in encoder.

Based on the token allocation from DFS, the encoder employs a two-stage attention structure for efficient processing. First, self-attention (MHA) performs dense relation modeling on the K_{sa} highest-scoring queries per level, enhancing features of high-confidence regions. Specifically, the score used to select these queries is defined as $S = S_{sem} \odot S_L$, which explicitly fuses a semantic score S_{sem} (derived from a lightweight classifier) with the geometry-aware foreground score S_L predicted by the HFG module. Subsequently, cross-level feature interaction is implemented via MSDA, enabling all selected tokens to aggregate contextual information from multiple scales and thus accomplish global enhancement and enrichment of features. Compared to processing all feature

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map tokens (complexity $\mathcal{O}(N^2)$), this design reduces computational complexity to $\mathcal{O}(K_{\text{total}}^2)$, significantly improving small object detection accuracy while controlling computational cost. Algorithm 1 summarizes our complete pipeline in encoder.

C. Foreground Confidence-Gated Attention

The decoder in standard Deformable DETR adopts a uniform cross-attention mechanism for all queries, overlooking their dynamic divergence during decoding: some queries have converged into clear target candidates (high confidence), while others remain in weak response states (low confidence). This undifferentiated treatment creates an optimization conflict: high confidence queries cannot leverage foreground priors for localization refinement, whereas low confidence queries may be misled by premature constraint from strong prior information, such as typical target shapes, textures, and positional distribution features learned during training, thus resulting in false matches.

To address this, we propose an FCGA module, which enables each query to adaptively modulate its reliance on foreground priors based on its own semantic confidence. This mechanism creates a dynamic balance: confident queries fully exploit spatial guidance for refinement, while uncertain ones preserve the necessary space for iterative feature updating to continuously optimize their representations and avoid being confined to erroneous states by strong priors.

First, for each query $\boldsymbol{q}_i$ (where $i \in \{1, 2, \ldots, N_q\}$ indexes the object queries in the decoder, with N_q denoting the total number of queries), the mechanism operates as follows. A lightweight MLP computes a semantic confidence score $g_i \in [0, 1]$, which acts as a dynamic gate

$$g_i = \text{sigmoid}\left(\text{MLP}\left(\boldsymbol{q}_i\right)\right). \tag{14}$$

Notably, g_i is explicitly designed to be distinct from the classification score. While the classification score predicts categorical identity, g_i measures the spatial convergence of the query to a target region. Then, query i computes its raw attention score S^i_{raw} based on its own features, representing its content-driven attention distribution. Following the standard scaled dot-product attention

$$S^i_{\text{raw}} = \frac{\boldsymbol{q}_i^T \cdot \text{Sample}\left(\mathbf{K}, \boldsymbol{s}_i\right)^T}{\sqrt{d}} \in \mathbb{R}^{L\times K} \tag{15}$$

where $\boldsymbol{q}_i \in \mathbb{R}^d$ denotes the feature embedding of the ith query, and $\mathbf{K} \in \mathbb{R}^{H\times W\times d}$ represents the multiscale key feature maps with spatial dimensions $H \times W$ and channel dimension d. $\boldsymbol{s}_i = \{\boldsymbol{s}_i^{(1)}, \ldots, \boldsymbol{s}_i^{(L\times K)}\}$ are the predicted deformable sampling locations for query i across L feature levels with K sampling points per level. $\text{Sample}(\mathbf{K}, \boldsymbol{s}_i) \in \mathbb{R}^{(L\times K)\times d}$ performs bilinear interpolation to extract key features at the specified locations (implemented via `F.grid_sample` in practice), and d is the channel dimension used for scaling.

Concurrently, query i samples the multilevel foreground score map $\mathcal{P}$ at its predicted deformable sampling locations to obtain a spatial guidance score P^i_{sampled}

$$P^i_{\text{sampled}} = \text{Sample}\left(\mathcal{P}, \boldsymbol{s}_i\right) \tag{16}$$

where $\mathcal{P} \in [0, 1]^{H\times W}$ is a multiscale foreground probability map indicating the likelihood of each spatial location belonging to a foreground object. $P^i_{\text{sampled}} \in [0, 1]^{L\times K}$ contains the sampled probability values at the $L \times K$ sampling locations for query i, ensuring dimensional consistency with S^i_{raw} for subsequent elementwise operations.

The final attention score S^i_{final} for query i is calculated by dynamically blending the raw scores with the spatially guided scores using the query-specific gate parameter g_i

$$S^i_{\text{final}} = (1 - g_i) \cdot S^i_{\text{raw}} + g_i \cdot \left(S^i_{\text{raw}} \odot P^i_{\text{sampled}}\right) \tag{17}$$

where g_i is the learnable scalar gate value, and $\odot$ denotes elementwise multiplication. This formulation elegantly allows a query with low confidence ($g_i \to 0$) to rely primarily on its data-driven raw attention S^i_{raw}, maintaining sufficient exploration of the feature space. Conversely, a high-confidence query ($g_i \to 1$) strongly incorporates the spatial prior by reweighting its raw attention with the foreground guidance from P^i_{sampled}, thus concentrating focus on the most promising regions.

Next, these fused scores are normalized via a Softmax function across all $L\times K$ sampling points for query i to produce the attention weights $\mathcal{A}_i$

$$\mathcal{A}_i = \text{Softmax}\left(S^i_{\text{final}}\right) \tag{18}$$

where $\mathcal{A}_i \in \mathbb{R}^{L\times K}$ represents the normalized attention weights that sum to 1 across all sampling points.

Finally, these weights are used to perform a weighted aggregation of the value features $\mathbf{V}$ sampled at the corresponding locations, yielding the final output feature $\boldsymbol{O}_i$ for query i

$$\boldsymbol{O}_i = \mathcal{A}_i \cdot \text{Sample}\left(\mathbf{V}, \boldsymbol{s}_i\right) \tag{19}$$

where $\mathbf{V} \in \mathbb{R}^{H\times W\times d}$ denotes the multiscale value feature maps with the same dimensions as the key features. $\text{Sample}(\mathbf{V}, \boldsymbol{s}_i) \in \mathbb{R}^{(L\times K)\times d}$ extracts value features at the sampling locations, and $\boldsymbol{O}_i \in \mathbb{R}^d$ is the aggregated output feature. This output encapsulates information from spatially relevant locations, weighted by the spatially calibrated attention mechanism, serving as the refined query representation.

The complete output for all queries is obtained by stacking the individual outputs: $\boldsymbol{O} = [\boldsymbol{O}_1, \ldots, \boldsymbol{O}_{N_q}]^T \in \mathbb{R}^{N_q\times d}$. This output is then processed following the standard Transformer decoder architecture.

In essence, FCGA introduces a self-supervised query-adaptive attention mechanism where gate parameters g_i are learned implicitly through end-to-end optimization. This mechanism enhances decoder robustness by preventing premature convergence of uncertain queries while accelerating refinement of confident predictions. The structure is illustrated in Fig. 4.

D. Loss Functions

Our method employs an end-to-end training strategy with a comprehensive loss function that supervises different aspects of oriented object detection. The total loss is formulated as follows:

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{cls}} + \mathcal{L}_{\text{bbox}} + \mathcal{L}_{\text{angle}} + \mathcal{L}_{\text{hf}}. \tag{20}$$

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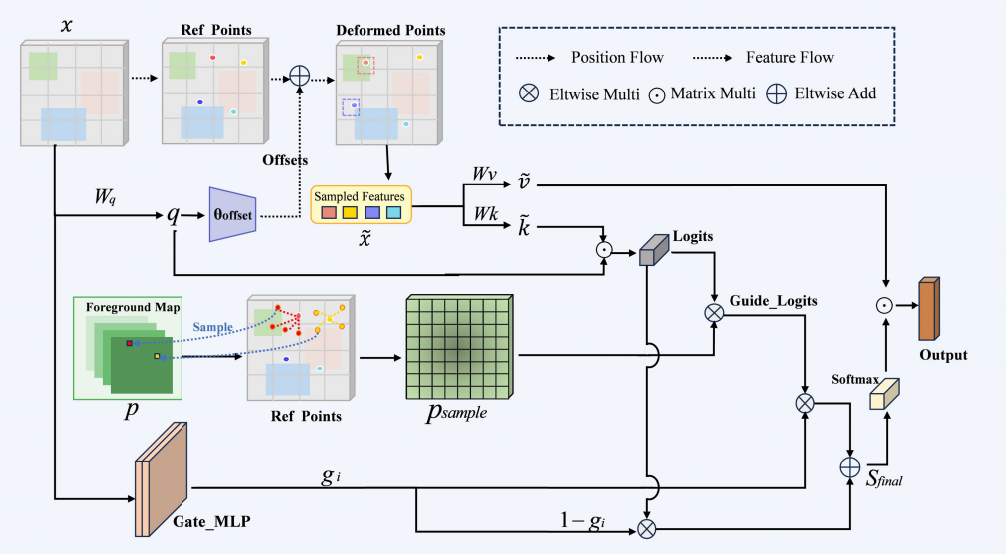


Fig. 4. Overview of the FCGA module. The module dynamically blends raw attention scores with spatial guidance from a foreground probability map through a learnable gate mechanism.

1) Classification Loss: To address the class imbalance problem inherent in object detection, we employ focal loss for classification

$$\mathcal{L}_{\text{cls}} = -\alpha\,(1-p_t)^{\gamma}\log(p_t) \tag{21}$$

where p_t is the predicted probability for the target class, $\alpha = 0.25$ is the balancing factor, and $\gamma = 2.0$ is the focusing parameter. The loss weight is set to 2.0.

2) Bounding Box Regression Loss: The bounding box regression loss combines L1 loss and generalized IoU (GIoU) loss to provide both coordinate-level and region-level supervision

$$\mathcal{L}_{\text{bbox}} = \lambda_{L1}\mathcal{L}_{L1}\left(b,\hat{b}\right) + \lambda_{\text{GIoU}}\mathcal{L}_{\text{GIoU}}\left(b,\hat{b}\right) \tag{22}$$

where b and $\hat{b}$ denote the ground truth and predicted bounding boxes in (x, y, w, h) format, respectively. The weights are set to $\lambda_{L1} = 2.0$ and $\lambda_{\text{GIoU}} = 5.0$. The bounding box loss is computed in the decoded coordinate space.

3) Angle Loss: For oriented object detection, we employ the aspect ratio sensitive loss from ARS-DETR [17]. The base angle loss is formulated as binary cross-entropy with sigmoid activation

$$\mathcal{L}_{\text{angle}} = -\sum_{i}\left[y_i\log\left(\sigma\left(\hat{\theta}_i\right)\right) + (1-y_i)\log\left(1-\sigma\left(\hat{\theta}_i\right)\right)\right]. \tag{23}$$

We then apply aspect ratio modulation (ARM) to adaptively weight the loss

$$\mathcal{L}_{\text{angle}}^{\text{ARM}} = \frac{2k_i}{1+k_i}\mathcal{L}_{\text{angle}} \tag{24}$$

where $k_i \geq 1$ is the aspect ratio of target i. The loss weight is set to 2.0.

4) Hierarchical Foreground Loss: To provide spatial guidance for the attention mechanism, we supervise the multiscale foreground probability maps using a focal loss

$$\mathcal{L}_{\text{hf}} = -\frac{1}{N}\sum_{i} w_i\left[F_{\text{gt},i}\log(p_i) + \left(1-F_{\text{gt},i}\right)\log(1-p_i)\right] \tag{25}$$

where $p_i = \sigma(F_i)$ is the predicted foreground probability, F_{gt} is the ground-truth foreground map, and the focal weight w_i is computed as follows:

$$w_i = \alpha_t\,(1-p_t)^{\gamma}, \quad p_t = \begin{cases} p_i, & \text{if } F_{\text{gt},i} = 1 \\ 1-p_i, & \text{otherwise} \end{cases} \tag{26}$$

with $\alpha = 0.25$ and $\gamma = 2.0$. This formulation emphasizes hard pixels near object boundaries while downweighting easy background regions. The loss weight is set to 10.0, enabling effective hierarchical foreground–background separation across multiple scales.

IV. Experiment

A. Datasets

1) DOTA-v1.0 [1]: The DOTA-v1.0 dataset contains 2806 aerial images (from 800 × 800 to 4000 × 4000 pixels) with 188 282 instances across 15 categories. We use both training and validation sets for training, and the test set for testing. We train our model on the combined training and validation sets, after cropping the images into 1024 × 1024 subimages with a 200 pixels overlap. We evaluate performance on the test set via the official DOTA server. The short names for categories are defined as follows: PL-Plane, BD-Baseball diamond, BR-Bridge,GTF-Ground field track, SV-Small vehicle, LV-Large vehicle, SH-Ship, TC-Tennis court, BC-Basketball court, ST-Storage tank, SBF-Soccer-ball field, RA-Roundabout, HA-Harbor, SP-Swimming pool, and HC-Helicopter.

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TABLE I

COMPARISONS WITH STATE-OF-THE-ART METHODS ON DOTA-V1.0 OBB TASK. THE RESULTS WITH RED AND BLUE COLORS INDICATE THE BEST AND SECOND-BEST RESULTS OF EACH COLUMN, RESPECTIVELY. ALL METHODS ARE OBTAINED UNDER SINGLE-SCALE TRAINING AND TESTING. "†" DENOTES OUR ADAPTATION FOR OBB DETECTION

Method	Backbone	PL	BD	BR	GTF	SV	LV	SH	TC	BC	ST	SBF	RA	HA	SP	HC	AP_{50}
CNN-based																	
KLD [34]	R-50	89.08	**84.18**	43.77	**72.33**	79.85	73.58	85.69	**90.88**	85.14	81.96	**65.86**	64.60	63.60	68.26	53.19	73.46
Oriented R-CNN [6]	R-50	89.46	82.12	54.78	70.86	78.93	83.00	88.20	90.67	**87.50**	84.68	63.97	67.69	74.94	68.84	52.28	75.87
S^2A-Net [32]	R-50	89.11	82.84	48.37	71.11	78.11	78.39	87.25	90.83	84.90	85.64	60.36	62.60	65.26	69.13	57.94	74.12
R^3Det [8]	R-50	**89.80**	83.77	48.11	66.77	78.76	**83.27**	87.84	90.82	85.38	85.51	**65.67**	62.68	67.53	78.56	**72.62**	76.47
QPDet [46]	R-50	**89.55**	83.66	54.06	73.93	78.93	83.08	88.29	90.89	86.60	84.80	62.03	65.55	74.16	70.09	58.16	76.25
TCD [47]	R-50	89.27	**83.79**	**56.91**	72.13	65.75	76.76	70.67	90.88	86.95	83.81	60.16	71.21	76.54	71.95	70.89	75.18
Oriented RepPoints [2]	R-50	87.02	83.17	54.13	71.16	80.18	78.40	87.28	**90.90**	85.97	**86.25**	59.90	**70.49**	73.53	72.27	58.97	75.97
SASM [48]	R-50	86.42	78.97	52.47	69.84	77.30	75.99	86.72	90.89	82.63	85.66	60.13	68.25	73.98	72.22	62.37	74.92
PSCD [35]	R-50	89.06	73.61	49.03	62.34	75.18	77.69	88.00	90.85	82.63	72.97	61.48	64.20	65.77	72.88	67.88	72.90
FCOSF [49]	R-50	88.43	78.59	48.04	65.17	79.80	80.24	87.08	90.90	82.75	84.65	58.39	66.44	64.66	71.88	57.60	73.64
RFpoint [50]	R-50	88.60	79.39	50.69	70.06	78.71	79.35	87.63	90.89	82.72	84.75	58.61	65.84	66.00	69.20	49.64	73.47
FRED [51]	R-50	89.37	82.12	50.84	73.89	77.58	77.38	87.51	90.82	86.30	84.25	62.54	65.10	72.65	69.55	63.41	75.56
SDANet [52]	R-50	89.12	83.59	52.99	73.08	78.48	79.97	87.22	90.64	85.20	83.37	63.11	66.70	76.65	70.13	57.07	75.82
O.RCNN [53]	R-50	89.46	82.12	54.78	70.86	78.93	83.00	88.20	90.90	**87.50**	84.68	63.97	67.69	74.94	68.84	52.28	75.87
HA-RDet [54]	R-50	88.98	83.34	55.02	72.04	78.25	81.43	86.94	90.91	85.46	85.48	61.33	61.57	74.50	68.97	56.86	75.41
Transformer-based																	
AO2-DETR [15]	R-50	87.99	79.46	45.74	66.64	78.90	73.90	73.30	90.40	80.55	85.89	55.19	63.62	51.83	70.15	60.04	70.91
EMO2-DETR [16]	Swin-T	89.03	79.59	48.71	60.23	77.34	76.42	84.53	90.77	84.80	85.68	48.86	67.55	66.32	71.54	53.49	72.32
Def-DETR(CSL) [33]	R-50	86.27	76.66	46.64	65.29	76.80	76.32	87.74	90.77	79.38	82.36	54.00	61.47	66.05	70.46	61.97	72.15
OrientedFormer [37]	R-50	88.14	79.13	51.96	67.34	81.02	83.26	88.29	90.90	85.57	**86.25**	60.84	66.36	73.81	71.23	56.49	75.37
RQFormer [55]	R-50	87.45	78.57	47.36	69.01	79.58	81.27	88.53	90.89	82.80	**86.21**	58.68	64.20	75.21	74.44	61.39	75.04
GSDet [56]	R-50	88.65	81.31	51.11	73.48	78.59	82.76	88.17	90.83	84.04	81.35	59.68	62.27	74.68	69.91	64.89	75.44
Salience-DETR† [21]	R-50	87.20	76.01	45.07	67.68	77.20	77.46	88.47	90.30	78.97	83.63	56.91	67.83	67.55	71.07	58.98	73.60
ARS-DETR [17]	R-50	86.97	75.56	48.32	69.20	77.92	77.94	87.69	90.50	77.31	82.86	60.28	64.58	74.88	71.76	**66.62**	74.16
ARS-DETR [17]	Swin-T	87.65	76.54	50.64	69.85	79.76	**83.91**	87.92	90.26	86.24	85.09	54.58	67.01	75.62	73.66	63.39	75.47
Ours																	
FCG-DETR	R-50	87.98	82.03	54.41	**72.41**	**80.78**	82.31	**88.92**	90.87	86.62	85.63	64.68	**70.89**	**76.77**	**74.28**	66.58	**76.86**
FCG-DETR	R-101	88.35	82.92	**54.83**	71.97	**80.75**	82.26	**88.81**	90.35	**87.05**	86.09	64.93	69.98	**78.07**	**75.04**	**66.82**	**77.51**

2) DOTA-v1.5 [1]: While DOTA-v1.5 shares its image set with DOTA-v1.0, it offers more comprehensive annotations. Key updates include the labeling of extremely small instances (under 10 pixels) and the addition of the "CC-Container crane" category, which brings the total instance count to 403 318.

3) DIOR-R [11]: This is a public dataset with OBB annotation. It is a large-scale collection of 23 463 aerial images, encompassing 1 92 518 instances across 20 common categories such as Airplane (APL), Ship (SH), and Vehicle (VE). The dataset is characterized by significant challenges, including high object size variability and diverse imaging conditions. Our experiments utilize both the training and validation sets for training and the test set for final performance evaluation.

B. Implementation Details and Evaluation Metrics

1) Implementation Details: Our models were implemented in PyTorch via the MMRotate [45] framework and trained using the AdamW optimizer with a mini-batch size of 2. The initial learning rate was set to 2×10^{-4}. The learning rate schedule consisted of a 1000-iteration warm-up phase, followed by a decay strategy where the rate was reduced by a factor of 10 at the 20th epoch. All experiments were conducted on a server equipped with 4 NVIDIA GeForce RTX 3090 GPUs.

2) Evaluation Metrics: The primary metrics for detection accuracy are the standard average precision (AP) benchmarks, including AP_{50}, AP_{75}, and $AP_{50:95}$. Model efficiency is evaluated by three metrics: model size (measured in Params), computational complexity (measured in GFLOPs), and training time (measured in epochs).

C. Comparison With the State-of-the-Art Methods

1) Results on DOTA-v1.0: We compare our method with state-of-the-art methods that employ diverse backbones in Table I. Our approach demonstrates superior performance on the DOTA-v1.0 dataset. With a ResNet-50 backbone and only 24 training epochs, FCG-DETR achieves 76.86% AP_{50}, outperforming most competitors. This represents a significant improvement over previous DETR-based methods: it outperforms Def-DETR(CSL) (72.15%) by 4.71 points, baseline ARS-DETR (74.16%) by 2.7 points, and the recently proposed RQFormer (75.04%) by 1.82 points. Compared with CNN-based detectors using a ResNet-50 backbone, our method also achieves competitive results, surpassing FRED (75.56%), SDANet (75.82%), O.RCNN (75.87%), and HA-RDet (75.41%) by considerable margins.

Notably, when using the same R-50 backbone, compared with ARS-DETR, our method achieves significant performance improvements on several challenging categories: ships (SH) with a gain of 1.23%, harbors (HA) with a gain of 1.89%, and swimming pools (SPs) with a gain of 7.52%. These targets face challenges such as complex backgrounds, variable scales, and dense distribution. Our proposed foreground dynamic token selection mechanism effectively allocates more computational resources to information-rich regions, thereby enabling better discrimination of small targets from complex backgrounds. By adopting ResNet-101, FCG-DETR further

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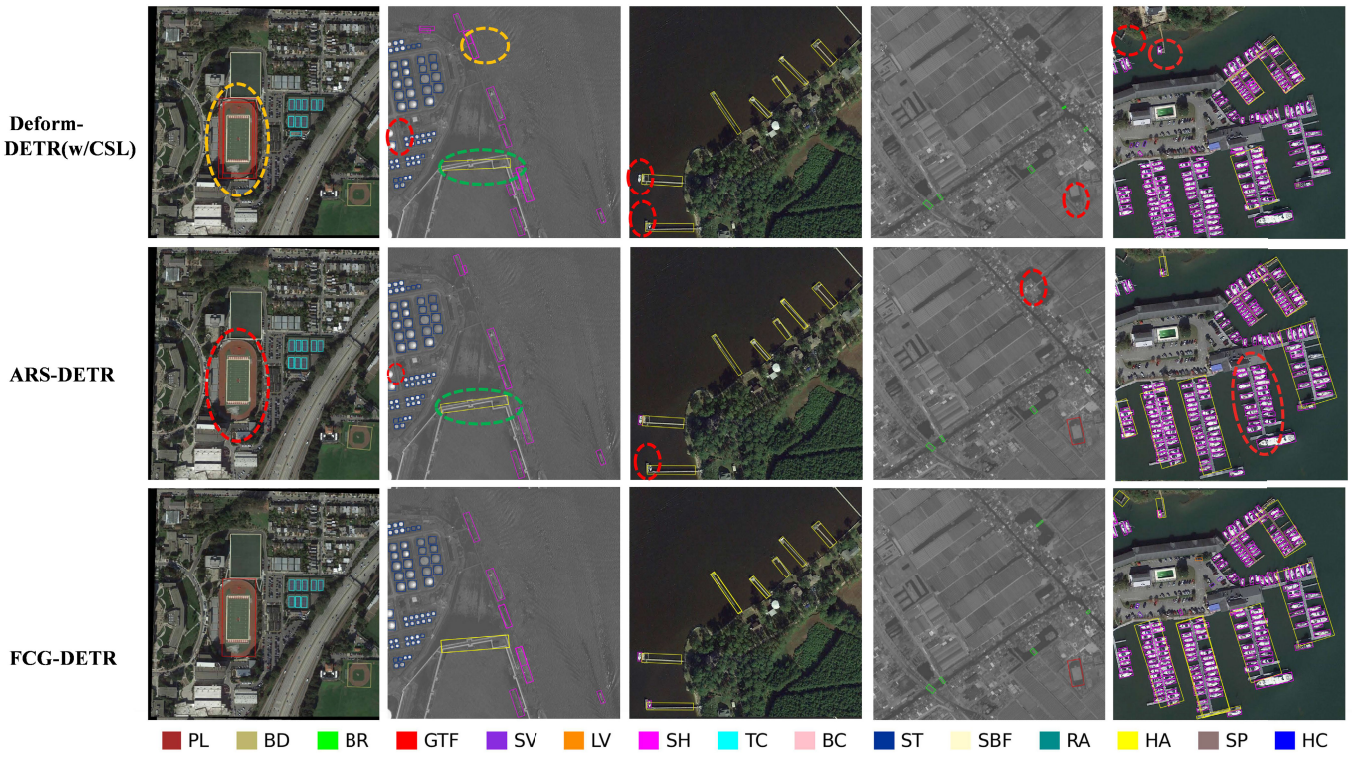


Fig. 5. Visualized detection results of Deform-DETR (w/CSL), ARS-DETR, and our FCG-DETR on the DOTA dataset with ResNet-50 as the backbone. Each color corresponds to an object class. The red dashed circles represent missed detections, the green ones indicate inaccurate bounding boxes, and the yellow ones denote false positives. The category abbreviations are defined as follows: PL: Plane, BD: Baseball diamond, BR: Bridge, GTF: Ground track field, SV: Small vehicle, LV: Large vehicle, SH: Ship, TC: Tennis court, BC: Basketball court, ST: Storage tank, SBF: Soccer-ball field, RA: Roundabout, HA: Harbor, SP: Swimming pool, HC: Helicopter.

TABLE II
MAIN RESULTS OF SMALL OBJECTS ON DOTA-V1.5. BEST PER COLUMN IN BOLD

Method	SV	LV	SH	ST	SP	CC	AP_{50}
Faster R-CNN-O [61]	51.28	68.98	79.37	67.50	65.28	1.54	62.00
Mask R-CNN [62]	51.31	71.34	79.75	66.07	64.47	0.96	62.67
HTC-O [63]	51.54	73.31	80.31	67.34	64.48	5.15	63.40
AO2-DETR [15]	55.34	74.50	79.57	77.58	73.11	**24.71**	66.26
OrientedFormer [37]	64.05	77.04	85.33	78.11	72.08	10.86	67.06
RQFormer [55]	62.16	**78.41**	88.94	77.83	72.69	13.37	67.43
FCG-DETR	**64.14**	78.13	**88.98**	**78.63**	**73.80**	12.78	**67.89**

improves to 77.51% AP_{50}, performing favorably against existing methods that adopt comparable training configurations.

Furthermore, our method maintains competitive accuracy on large-scale targets such as ground track fields (GTFs: 72.41%) and storage tanks (STs: 85.63%). This consistent performance across varying scales demonstrates the integration of our three core modules. The HFG module constructs representations that conform to the geometric morphology of objects through multiscale feature extraction, the DFS adaptively concentrates resources on informative regions (particularly benefiting small and dense objects), and the FCGA modulates decoder queries to emphasize foreground while suppressing background interference. These designs enable FCG-DETR to achieve balanced performance across the full spectrum of object scales, ensuring precise small object localization while maintaining complete large object representation.

Fig. 5 compares the detection results of three methods on the DOTA dataset, revealing distinct failure patterns and our method's advantages. Deform-DETR (w/CSL) suffers from severe missed detections (red circles) on small and densely distributed objects, particularly struggling with harbor structures (column 2) and small vehicles (column 4), indicating insufficient attention to small-scale foreground regions. ARS-DETR improves recall but exhibits inaccurate localization (green circles), particularly for elongated objects such as bridges (column 3) and large vehicles (column 1), suggesting limitations in capturing precise OBBs for objects with extreme aspect ratios. In contrast, our FCG-DETR achieves more complete detection coverage while maintaining precise localization across diverse object scales and orientations.

2) Results on DOTA-v1.5: Table II presents results on DOTA-v1.5, a challenging benchmark with extremely small objects. Our FCG-DETR achieves 67.89% AP_{50}, outperforming both CNN-based and transformer-based approaches. Compared to AO2-DETR (66.26%), our method gains 1.63 points. Against recent OrientedFormer (67.06%), we achieve a 0.83-point improvement with particularly strong performance on small object categories: SH (88.98% versus 85.33%), SV (64.14% versus 64.05%), and SP (73.80% versus 72.08%). These results confirm that our DFS strategy effectively captures multiscale contextual information for oriented object detection.

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TABLE III

COMPARISON WITH STATE-OF-THE-ART METHODS ON DIOR-R. THE RESULTS WITH RED AND BLUE COLORS INDICATE THE BEST AND SECOND-BEST RESULTS. "†" DENOTES OUR ADAPTATION FOR OBB DETECTION

Method	Backbone	APL	APO	BF	BC	BR	CH	DAM	ETS	ESA	GF	GTF	HA	OP	SH	STA	STO	TC	TS	VE	WM	AP_{50}
CNN-based																						
Oriented Rep [2]	R-50	**70.03**	46.11	76.12	87.19	39.14	**78.76**	34.57	71.80	80.42	76.16	79.41	45.48	54.90	87.82	**77.03**	68.07	81.60	56.83	51.57	**71.25**	66.71
DCFL [57]	R-50	68.60	53.10	76.70	87.10	42.10	**78.60**	34.50	71.50	**80.80**	**79.70**	79.50	47.30	**57.40**	85.20	64.60	66.40	81.50	58.90	50.90	70.90	66.80
DFDet [6]	R-50	61.92	38.83	**77.41**	81.36	34.11	74.97	26.26	62.31	76.06	75.56	79.62	38.26	52.76	80.40	73.11	68.27	81.38	52.23	44.11	63.35	62.11
Gliding Vertex [58]	R-50	65.35	28.87	74.96	81.33	33.88	74.31	19.58	70.72	64.70	72.30	78.68	37.22	49.64	80.22	69.26	61.13	81.49	44.76	47.71	65.04	60.06
AOPG [59]	R-50	62.39	37.79	71.62	**87.63**	40.90	72.47	31.08	65.42	77.99	73.20	**81.94**	42.32	54.45	81.17	72.69	71.31	81.49	60.04	53.38	69.99	64.41
DODet [60]	R-50	63.40	43.35	72.11	81.32	**43.12**	72.59	33.32	**78.77**	70.84	74.15	75.47	**48.00**	**59.31**	85.41	74.04	**71.56**	81.52	55.47	51.86	66.40	65.10
QPDet [46]	R-50	63.22	41.39	71.97	**88.55**	41.23	72.63	28.82	**78.90**	69.00	70.07	**83.01**	47.83	55.54	81.23	72.15	62.66	**89.05**	58.09	43.38	65.36	64.20
Transformer-based																						
OrientedFormer [37]	R-50	65.65	48.69	**78.79**	87.17	41.90	76.34	34.37	72.14	81.40	75.34	79.83	45.15	56.12	**88.66**	67.59	**72.68**	**87.32**	**60.31**	**56.54**	**70.56**	67.28
RQFormer [55]	R-50	66.71	**54.14**	73.11	81.80	45.25	77.60	33.37	70.09	79.18	72.24	75.42	47.18	58.27	**89.50**	70.98	74.62	82.83	55.03	56.23	66.61	66.51
Salience-DETR† [21]	R-50	64.07	44.51	72.83	80.92	41.07	74.43	29.80	77.50	66.07	75.16	76.13	45.80	56.93	86.17	70.98	70.83	83.64	54.71	51.81	64.93	65.27
ARS-DETR [17]	R-50	68.00	**54.17**	74.43	81.65	41.13	75.66	**34.89**	73.07	81.92	76.10	78.62	36.33	55.41	84.55	70.09	72.23	81.14	**61.52**	50.57	70.28	66.12
Ours																						
FCG-DETR	R-50	**68.17**	53.31	73.00	86.91	**45.25**	75.02	**34.96**	74.24	**81.92**	**77.10**	78.32	**48.00**	54.86	85.50	**73.32**	69.69	86.20	60.04	**56.23**	68.47	**67.85**

3) Results on DIOR-R: Table III evaluates performance on DIOR-R, a fine-grained remote sensing dataset with 20 object categories. Our FCG-DETR achieves 67.85% AP_{50}, surpassing all baseline methods and establishing state-of-the-art performance. Compared to CNN-based detectors, FCG-DETR significantly outperforms the strongest baseline Oriented Rep (66.71%) by 1.14 points and DCFL (66.80%) by 1.05 points. Against transformer-based methods, we exceed OrientedFormer (67.28%) by 0.57 points, RQFormer (66.51%) by 1.34 points, and ARS-DETR (66.12%) by 1.73 points, demonstrating the effectiveness of our method. FCG-DETR shows balanced capability across diverse object types. For small dense objects like bridge (BR: 45.25%) and harbor (HA: 48.00%), our method ranks first, validating enhanced feature discrimination for densely distributed instances. For large structured objects like ship (SH: 89.50%) and expressway service area (ESA: 81.92%), FCG-DETR achieves top performance, demonstrating robust feature representation for large-scale objects. For dam (DAM: 34.96%) and vehicle (VE: 56.23%), we obtain best and second-best results respectively, showing improved detection for objects with complex backgrounds.

Notably, while certain CNN-based methods excel on specific categories (e.g., Oriented Rep on airplane APL: 70.03%, wind mill WM: 71.25%), FCG-DETR maintains more balanced performance across all 20 categories without significant weaknesses. This validates that our method adapts to multiscale and multicategory-oriented objects in complex remote sensing scenarios.

D. Ablation Study

Ablation studies are conducted on the DOTA-v1.0 dataset, with ARS-DETR as the base framework and ResNet-50 as the backbone. We incrementally integrate our three proposed modules (HFG, DFS, and FCGA) to evaluate their individual and collective contributions. The results are detailed in Table IV, with checkmarks (✓) denoting module inclusion.

1) Main Component Analysis: We systematically integrate the proposed modules into the ARS-DETR baseline to validate their effectiveness. Starting with HFG alone, which generates multiscale probability foreground maps, AP_{50} reaches 74.16%. Adding DFS improves AP_{50} to 75.17% (+1.01%), demonstrating that dynamic token selection effectively eliminates background tokens while preserving foreground information. Incorporating FCGA with HFG yields AP_{50} of 75.85% (+1.69%), showing that enhanced feature representation through fine-grained spatial–channel interactions is more impactful than token selection alone for discriminating small and densely distributed objects.

TABLE IV

EFFECTIVENESS OF INDIVIDUAL MODULES EXPERIMENTAL RESULTS OF EACH PROPOSED COMPONENT ON DOTA-V1.0 DATASET WITH R-50 BACKBONE

Methods	HFG	DFS	FCGA	AP_{50}	AP_{75}	$AP_{50:95}$
FCG-DETR				74.16	47.87	45.60
	✓	✓		75.17	48.09	46.08
	✓		✓	75.85	48.27	46.63
	✓	✓	✓	76.86	48.49	47.11

TABLE V

ABLATION STUDY ON DIFFERENT TOKEN SELECTION METHODS AND THE TOTAL NUMBER OF TOKENS (K_{TOTAL}). THE INPUT IMAGE RESOLUTION IS 1024×1024

Method	Layer Ratio	K_{total}	AP_{50}	AP_{75}	$AP_{50:95}$	FLOPs
ARS -DETR	$(1, 1, 1, 1)$	21760	74.16	47.87	45.60	206G
Salience -DETR	$(.4, .8, 1, 1)$	11109	73.60	47.67	45.08	168G
FCG -DETR (w/o FCGA)	-	14000	74.97	47.81	46.28	173G
	-	12000	75.05	47.95	46.12	168G
	-	**10000**	**75.17**	**48.09**	**46.08**	**163G**
	-	8000	74.82	47.78	45.79	158G
	-	6000	74.30	47.53	45.34	154G

Combining all three modules achieves AP_{50} of 76.86%, a substantial 2.70% improvement over baseline. This gain reflects the complementary nature of the modules: DFS provides refined token sets that enable FCGA to perform more effective feature interactions. Consistent improvements across all metrics (AP_{75}: +0.62%, $AP_{50:95}$: +1.51%) validate that our modules enhance both detection recall and localization precision for oriented object detection in remote sensing images.

2) Comparison of Different Number of Total Token in Encoder: Table V presents an ablation study on the total number of encoder tokens K_{total}. The results reveal that

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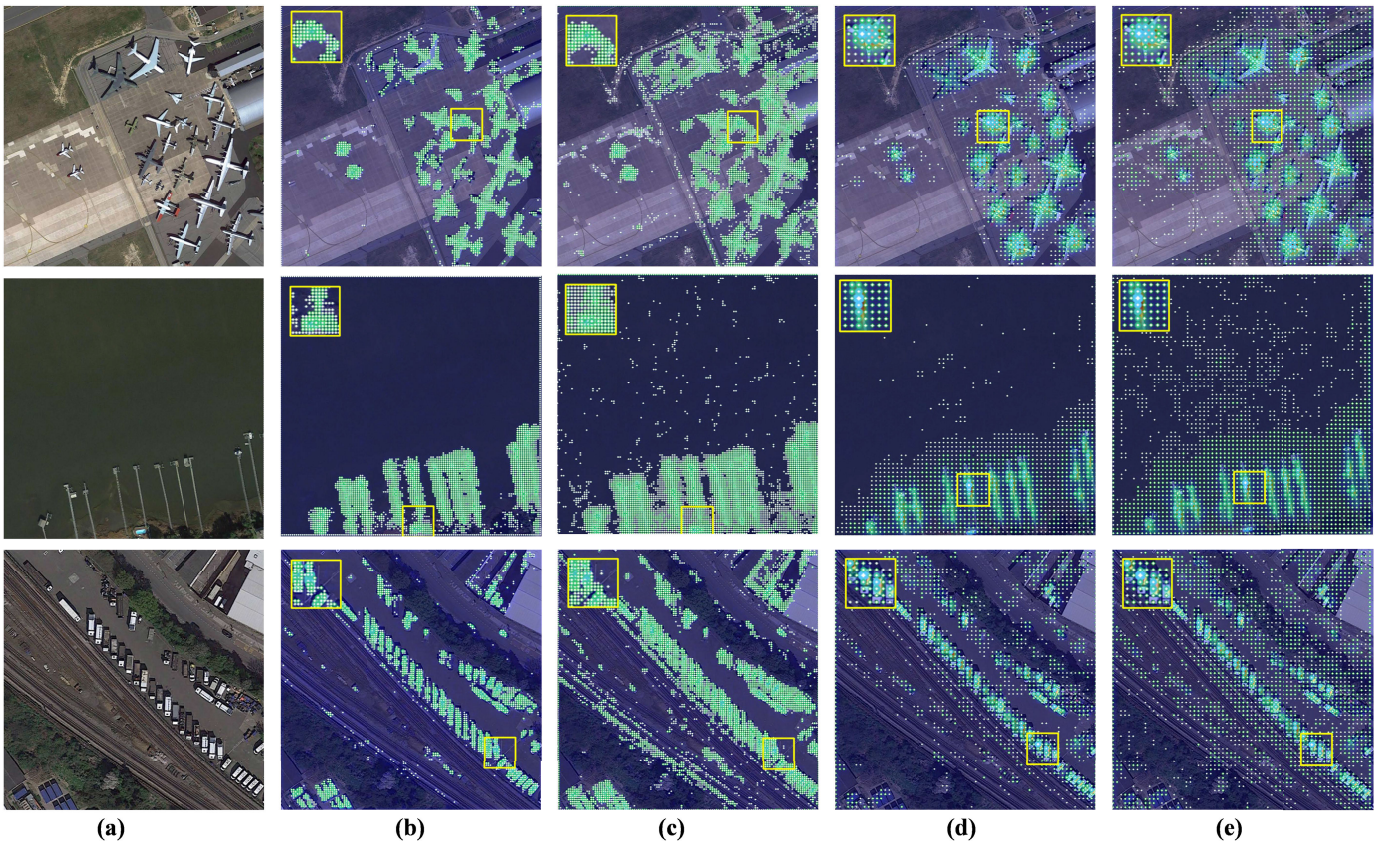


Fig. 6. Visualization of DFS across encoder layers on DOTA-v1.0. (a) Original image. (b)–(e) Selected tokens at layers S_2, S_3, S_4, and S_5 respectively. The blue dots indicate the positions with high scores, and the yellow magnified boxes represent the sampling points of the local regions centered on them.

performance peaks at $K_{\text{total}} = 10\,000$ with 75.17 AP_{50}, surpassing both the full-token ARS-DETR (74.16) and configurations with more tokens (such as 75.05 at 12 000 and 74.97 at 14 000). This nonmonotonic trend indicates that an optimal token number exists: since objects typically occupy only a sparse fraction of the image, forcing the model to sample excessive tokens inevitably incorporates complex background clutter, which dilutes the focus of the self-attention mechanism. Conversely, overly aggressive token reduction (≤ 8000) causes critical information loss for densely packed small targets, directly leading to false negatives. Notably, our DFS method outperforms Salience DETR's fixed ratio strategy by 1.57 AP_{50} (75.17 versus 73.60) despite using 10% fewer tokens (10 000 versus 11 109), confirming that variance-guided dynamic allocation is more effective than static layer ratios, as it can flexibly adapt the token numbers across different feature levels based on actual scene complexity and object scales, rather than relying on rigid rules. The optimal configuration achieves 21% FLOPs reduction compared to ARS-DETR (163 versus 206 G) while maintaining superior accuracy, making it suitable for resource-constrained deployment.

Fig. 6 visualizes the hierarchical token selection process across encoder layers, revealing how our DFS adapts to the multiscale nature of remote sensing objects. In the shallow layer S_2, tokens exhibit dense and uniform distribution with fine-grained spatial coverage, capturing low-level features essential for detecting small objects like individual aircraft and vehicles. As the network deepens to S_3 and S_4, token allocation becomes progressively sparser and more selective, concentrating on semantically rich regions where objects cluster (e.g., aircraft parking areas, harbor zones with dense ships, and vehicle-packed roads). The color intensity variation indicates that DFS assigns higher importance scores to tokens surrounding actual objects while suppressing background regions. In the deepest layer S_5, token distribution reaches maximum sparsity with strong focus on high-level semantic regions, effectively filtering out vast empty areas characteristic of aerial imagery.

TABLE VI

ABLATION STUDY ON $w_{\mathcal{I}}$ AND $w_{\mathcal{D}}$ FOR DFS. CAP TYPE DENOTES CAPACITY WEIGHTING MECHANISM. INPUT SIZE 1024×1024, $K_{\text{TOTAL}} = 10\,000$

$w_{\mathcal{I}}$	$w_{\mathcal{D}}$	Cap Type	AP_{50}
-	-	-	74.16
1.0	0.0	Log	74.29
0.0	1.0	Log	73.56
0.7	0.3	Log	73.81
0.5	**0.5**	**Log**	**75.17**
0.5	0.5	Linear	73.10
0.5	0.5	Sqrt	74.45

3) Comparison of Different Information Density Weight $w_{\mathcal{I}}$ and Relative Dispersion Weight $w_{\mathcal{D}}$: Table VI validates

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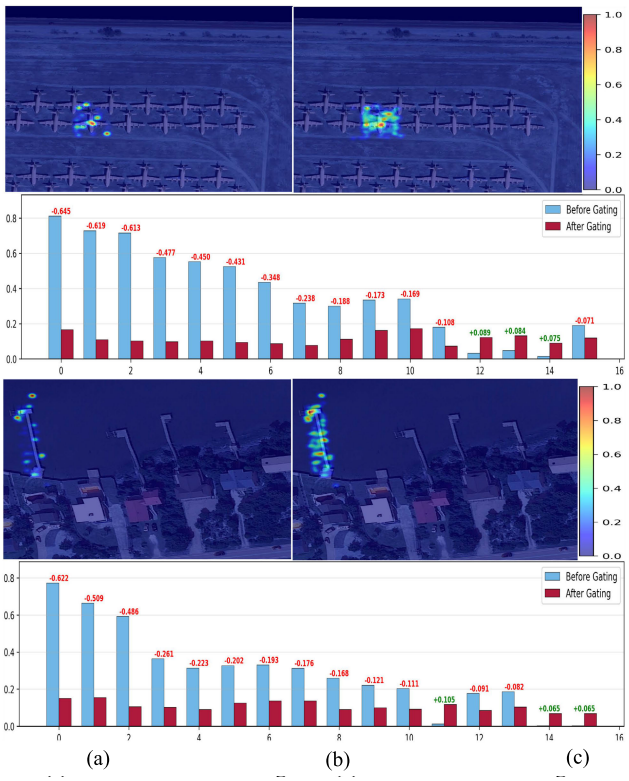


Fig. 7. (Top and bottom rows) Visualization of gating mechanism effects on attention distribution across two different scenarios. (a) Attention before gating. (b) Attention after gating. (c) Attention weight changes (top 16 sample points).

our design choices through systematic ablation. The balanced fusion ($w_{\mathcal{I}} = 0.5$, $w_{\mathcal{D}} = 0.5$) with logarithmic capacity weighting achieves optimal performance (75.17 AP_{50}), substantially outperforming single-metric approaches: information density alone (74.29 AP_{50}) and dispersion alone (73.56 AP_{50}). This 0.88 and 1.61 AP_{50} improvement demonstrates that the two metrics capture complementary spatial characteristics for remote sensing imagery. Information density identifies regions with concentrated object distributions (e.g., harbor zones with dense ships and urban areas with clustered vehicles), while dispersion ensures adequate token coverage across sparse backgrounds where isolated large objects (e.g., individual aircraft and storage tanks) may appear. The fusion of these two metrics directly addresses the core challenge of remote sensing scenarios: critical targets may either be densely clustered in local hotspots or sparsely distributed across vast empty regions. This design not only avoids insufficient token allocation for isolated yet vital objects in sparse areas, but also prevents excessive token assignment to redundant backgrounds in dense but low-informative regions, ultimately achieving an optimal balance in detection performance.

E. Effects of FCGA Module

Fig. 7 demonstrates the attention refinement capability of our gating mechanism across diverse challenging scenarios. Specifically, the attention distribution is relatively dispersed across multiple regions (Fig. 7(a) before gating), with attention weights scattered among background elements. After applying the gating mechanism with a high coefficient [Fig. 7(b)], the attention becomes highly concentrated on the target. For instance, in the densely packed airport scene, the mechanism accurately activates on the specific target aircraft without being disturbed by adjacent planes. Similarly, in the harbors scene, guided by the foreground prior, the attention precisely aggregates along the shape of the targets despite their exceptionally large aspect ratios. Both cases effectively filter out irrelevant background information. Quantitative analysis in Fig. 7(c) reveals the consistent selective filtering behavior: taking the airport scene as an example, among the top-16 most-changed sampling points, 13 points experience significant suppression with reduction magnitudes ranging from −0.071 to −0.645, while three points show enhancement ranging from +0.075 to +0.089.

These results clearly demonstrate that the gating mechanism acts as an intelligent attention gate, predominantly suppressing background noise while preserving object relevant features, regardless of target density or shape.

Fig. 8 demonstrates the evolution of deformable attention sampling patterns across different decoder layers. As the decoder depth increases, the sampling points around each query reference point gradually concentrate and align with the target objects. The predicted bounding boxes also progressively approach the ground-truth (GT) object boundaries. This phenomenon demonstrates the decoder's capability for iterative learning and adaptive adjustment.

F. Convergence and Training Epochs

To further examine whether our token selection mechanism affects model convergence, we plot the training curves of different methods in Fig 9. FCG-DETR (green solid line) demonstrates faster convergence speed in the early training stages, rapidly achieving high performance levels within the first six epochs and maintaining a leading position throughout the entire training process. This acceleration is attributed to the explicit foreground priors, which guide the model to focus on informative target regions from the very beginning. In contrast, although ARS-DETR (orange solid line) achieves comparable final performance, its convergence speed is significantly slower. Combined with the data analysis from Table V, FCG-DETR without the FCGA module achieves 46.08 $AP_{50:95}$ performance when $K_{\text{total}} = 10\,000$, requiring only 163 G FLOPs, which represents approximately 21% reduction in computational cost compared to ARS-DETR's 206 G FLOPs. This demonstrates that FCG-DETR's token selection strategy is more efficient, capable of preserving critical semantic information under a smaller token number.

G. Comparison of GFLOPs and Params

As shown in Fig. 10, ARS-DETR has the highest computational cost, reaching 206 GFLOPs, while Deform-DETR w CSL stands at 203 GFLOPs, with both being quite close to each other. In contrast, FCG-DETR's computational complexity is significantly reduced to 168 GFLOPs, representing approximately an 18% reduction compared to the other two methods, demonstrating superior computational

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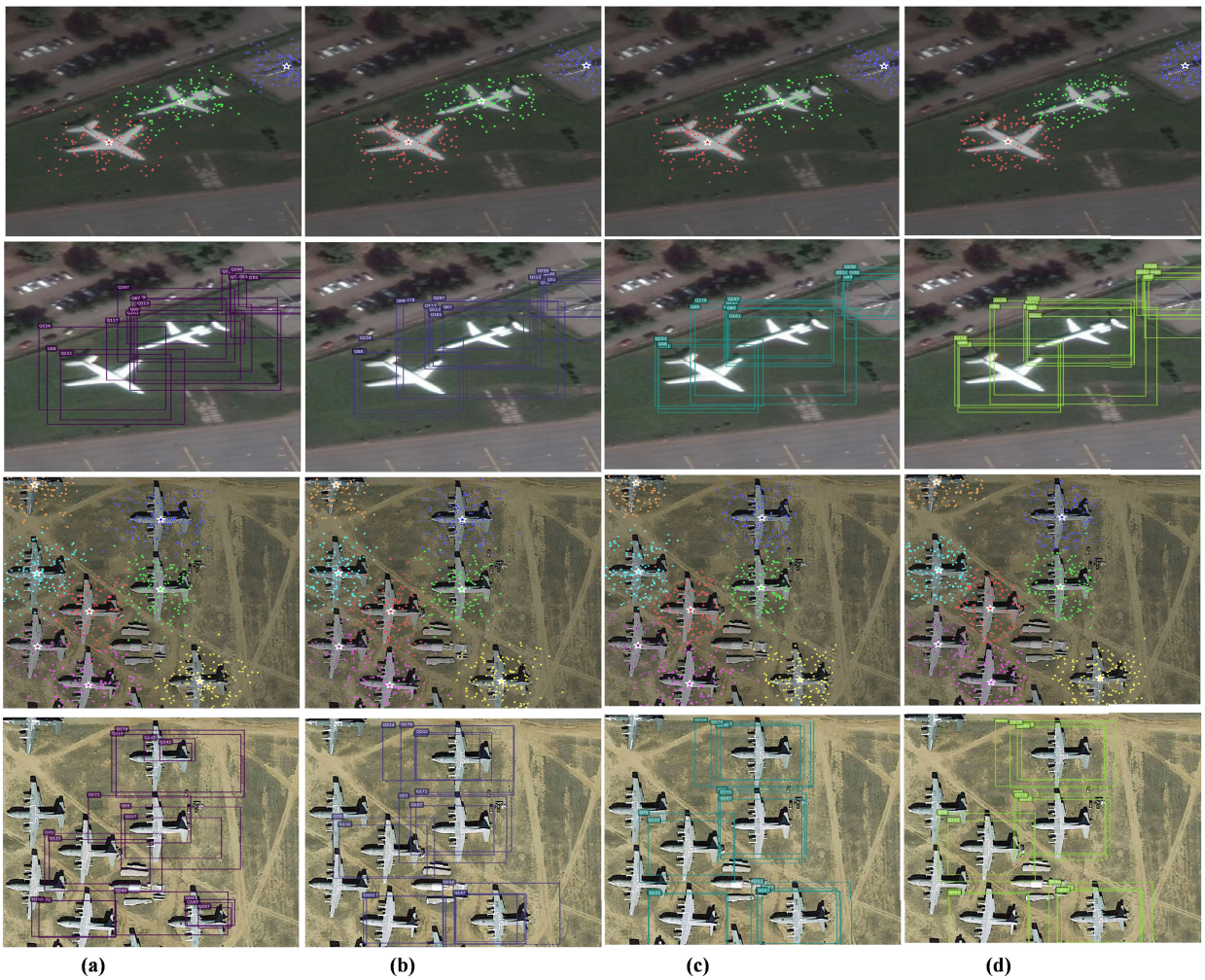


Fig. 8. Visualization of deformable attention sampling mechanisms across decoder layers (layers 1, 3, 4, and 6). Star markers represent reference points for object queries, colored dots represent sampling points learned by the model, and colored boxes represent detection boxes output by the decoder. (a) Layer 1. (b) Layer 3. (c) Layer 4. (d) Layer 6.

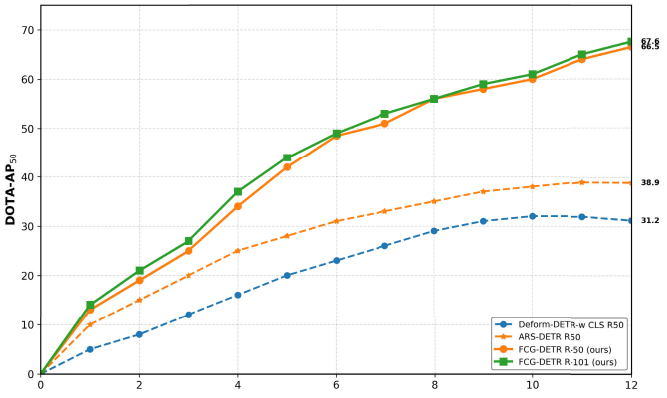


Fig. 9. Convergence curves of deformable DETR with CLS, ARS-DETR, and our method on DOTA-v1.0.

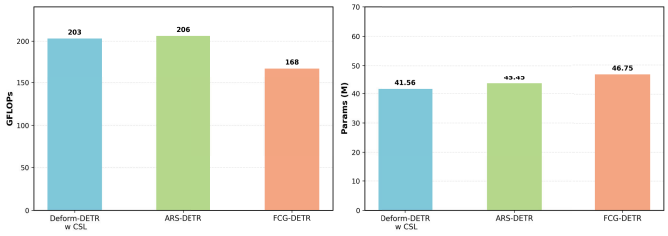


Fig. 10. Comparison of computational cost and parameters across different DETR methods.

efficiency. FCG-DETR has the highest number of parameters (46.75 M) because it incorporates additional foreground generation heads and gating heads, which are lightweight network structures. Despite having slightly more parameters, the substantial reduction in computational complexity and performance improvement indicates that this model achieves an effective balance between accuracy and efficiency, making it more advantageous in practical application scenarios.

H. Suboptimal Results Analysis

Despite competitive performance, FCG-DETR exhibits suboptimal results in extreme scenarios (Fig. 11). False positives occasionally arise from severe weather interferences like thick

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Fig. 11. Visualizations of suboptimal results. (Top row) Detection results of our FCG-DETR. (Bottom row) GT.

clouds (Left), or high visual similarity between backgrounds and targets (Middle). Furthermore, although our HFG module mathematically normalizes aspect ratios, extremely elongated targets (e.g., the narrow harbor) still cause false negatives (Right). This occurs because the exponential decay of the Gaussian prior yields insufficient supervision signals at the extreme ends, hindering effective feature aggregation.

V. Conclusion

In this article, we propose FCG-DETR, a novel transformer-based framework for oriented object detection. By integrating a probabilistic foreground guidance mechanism, dynamic token selection, and confidence-gated attention, FCG-DETR effectively distinguishes targets from complex backgrounds while adaptively optimizing computational resource allocation. Extensive experiments on multiple benchmarks demonstrate that our method achieves state-of-the-art performance.

However, the proposed method faces specific limitations in extreme scenarios. First, it struggles with extreme geometric scales: ultrasmall targets suffer from feature dilution below the downsampling stride, and extremely elongated targets lack sufficient extremity supervision due to the Gaussian prior's decay. Second, while our token selection is efficient, applying this Transformer architecture to massive high-resolution aerial imagery still incurs latency, posing challenges for strict real-time deployment. Future work will explore lightweight network designs (e.g., model compression) and extend this probabilistic framework to resource constrained tasks.

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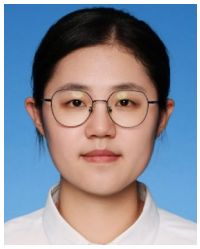
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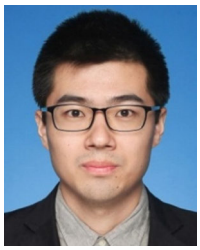
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